Algebraic Geometry – Exercise Sheet 3

Week 3: Schemes, morphisms, gluing and first global properties

Monday and Thursday afternoon sessions; Lectures 11–15

Difficulty. $\star$ routine; $\star\star$ synthesis; $\star\star\star$ challenge or extension.
Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

Monday afternoon — material available: Lectures 1–11

Exercise 1. Global functions as morphisms to the affine line $\star$ CORE

Let X be a scheme.

(i) Explain why a global section $s \in \Gamma(X, \mathcal{O}_X)$ determines a morphism

$$f_s : X \longrightarrow \mathbb{A}^1_{\mathbb{Z}}.$$

(ii) Show that

$$\mathrm{Hom}(X, \mathbb{A}^1_{\mathbb{Z}}) \simeq \Gamma(X, \mathcal{O}_X).$$

(iii) If $X = \mathrm{Spec}(A)$, describe the morphism associated with $a \in A$.

Exercise 2. The reduced point and the double point $\star$ CORE

Inside $\mathbb{A}^1_{\Bbbk} = \mathrm{Spec}(\Bbbk[t])$, consider

$$P = \mathrm{Spec}(\Bbbk[t]/(t)), \qquad D = \mathrm{Spec}(\Bbbk[t]/(t^2)).$$

(i) Show that the two closed immersions have the same image as subsets of $\mathbb{A}^1_{\Bbbk}$.

(ii) Show that P and D are not isomorphic as schemes.

(iii) Compute the maps on local rings at their unique points.

Exercise 3. A morphism between affine schemes $\star$ CORE

Let

$$\varphi : \Bbbk[u, v] \longrightarrow \Bbbk[t], \qquad u \longmapsto t, \quad v \longmapsto t^2.$$

(i) Describe the induced morphism $f : \mathbb{A}^1_{\Bbbk} \to \mathbb{A}^2_{\Bbbk}$.

(ii) Determine $\ker(\varphi)$ and factor f through a closed subscheme of $\mathbb{A}^2_{\Bbbk}$.

(iii) Show that the resulting morphism onto that closed subscheme is an isomorphism.

Exercise 4. An arithmetic spectrum $\star\star$ CHOICE

Let $A = \mathbb{Z}[s]/(s^2 - 3)$. Describe the inverse image in $\mathrm{Spec}(A)$ of the points (2), (3), and (p) for $p \neq 2, 3$, according to the factorisation of $s^2 - 3$ modulo p.

Thursday afternoon — material available: Lectures 1–14

Exercise 5. Dual numbers and tangent vectors $\star\star$ CORE

Let $D = \mathrm{Spec}(\Bbbk[\varepsilon]/(\varepsilon^2))$ and let $p = (a, b) \in \mathbb{A}^2_{\Bbbk}(\Bbbk)$.

(i) Show that morphisms $D \to \mathbb{A}^2_{\Bbbk}$ whose closed point maps to p are uniquely of the form

$$x \longmapsto a + u\varepsilon, \qquad y \longmapsto b + v\varepsilon.$$

(ii) Let $C = \mathbb{V}(f) \subset \mathbb{A}^2_{\Bbbk}$. Show that such a morphism factors through C precisely when

$$f(a, b) = 0, \qquad f_x(a, b)u + f_y(a, b)v = 0.$$

(iii) Apply this to the parabola $y = x^2$.

Exercise 6. The line with two origins ⋆⋆ CORE

Glue two copies

$$X_1 = \mathrm{Spec}(\Bbbk[t_1]), \qquad X_2 = \mathrm{Spec}(\Bbbk[t_2])$$

along $D(t_1) \simeq D(t_2)$ by $t_1 \mapsto t_2$. Let X be the resulting scheme.

(i) Describe the underlying topological space.

(ii) Show that $\Gamma(X, \mathcal{O}_X) \simeq \Bbbk[t]$.

(iii) Show that the canonical map $X \to \mathrm{Spec}\,\Gamma(X, \mathcal{O}_X)$ identifies the two origins.

(iv) Deduce that X is not affine.

Exercise 7. Global functions on the projective line ⋆⋆ CHOICE

Construct $\mathbb{P}^1_{\Bbbk}$ by gluing $U_0 = \mathrm{Spec}(\Bbbk[t])$ and $U_1 = \mathrm{Spec}(\Bbbk[s])$ along $s = t^{-1}$. Prove that

$$\Gamma(\mathbb{P}^1_{\Bbbk}, \mathcal{O}_{\mathbb{P}^1}) = \Bbbk.$$

Exercise 8. The projective conic from its standard opens ⋆⋆ CORE

Let

$$S = \Bbbk[x_0, x_1, x_2]/(x_0x_2 - x_1^2), \qquad C = \mathrm{Proj}(S).$$

(i) Compute $(S_{x_0})_0$ and $(S_{x_2})_0$.

(ii) Identify $D_+(x_0)$ and $D_+(x_2)$ with affine lines.

(iii) Describe their overlap and recover the gluing rule for $\mathbb{P}^1$.

Exercise 9. Connectedness and idempotents ⋆⋆ CORE

(i) Prove that $\mathrm{Spec}(A)$ is disconnected if and only if A has a non-trivial idempotent.

(ii) Apply this to $A = \Bbbk[t]/(t^2 - t)$.

(iii) Show that $\mathrm{Spec}(\Bbbk[x, y]/(xy))$ is connected although reducible.

Exercise 10. Reduction and its universal property ⋆⋆ CORE

Let

$$A = \Bbbk[x, y]/(x^2, xy), \qquad X = \mathrm{Spec}(A).$$

(i) Compute the nilradical and X_{red}.

(ii) Let R be a reduced $\Bbbk$-algebra. Show that every homomorphism $A \to R$ factors uniquely through A_{red}.

(iii) Translate this into a statement about morphisms of schemes.

Exercise 11. A weighted projective chart ⋆⋆⋆ OPTIONAL

Let $R = \Bbbk[x, y, z]$ with $\deg x = \deg y = 1$ and $\deg z = 2$, and set $X = \mathrm{Proj}(R)$. Show that $D_+(x)$ and $D_+(y)$ are affine planes and that

$$D_+(z) \simeq \mathrm{Spec}\big(\Bbbk[a, b, c]/(ac - b^2)\big).$$

Identify the singular point of this chart.

After Friday's lecture or preparation for Week 4

Exercise 12. Components and generic points ⋆⋆ CHOICE

Let

$$I = (xy, xz) \subset \Bbbk[x, y, z], \qquad X = \mathrm{Spec}(\Bbbk[x, y, z]/I).$$

Show that $I = (x) \cap (y, z)$. Determine the irreducible components, their generic points and dimensions, and decide whether X is connected or integral.

Exercise 13. The function field of the cusp ⋆⋆ CHOICE

Let

$$C = \mathrm{Spec}\big(\Bbbk[x, y]/(y^2 - x^3)\big).$$

Show that C is integral and that $K(C) \simeq \Bbbk(t)$.